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Assignment - 1

1. Differences between static physical model and dynamic physical model with example.

⇒ The differences between static physical model and dynamic physical model are given below:

Static Physical Model	Dynamic Physical Model
i) Static physical model is a scaled down model of a system which does not change with time.	i) Dynamic physical model is one which change with time or which are function of time.
ii) It is used to understand structure or shape of a system.	ii) It is used to understand behavior and operation of a system.
iii) It can't simulate real-time behavior.	iii) It can simulate real-time behavior.
iv) User cannot interact with it meaningfully.	iv) User can interact and observe response.
v) It is rarely used for simulation.	v) It is widely used for simulation.
vi) Examples: map, blueprint, statue, etc.	vi) Examples: flight simulator, weather model, traffic simulator etc.

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2. Define simulation. What are various steps involved in simulation.

⇒ Simulation is the process of imitating the operations of various kinds of real world facilities or process.

The various steps involved in simulation are given below:

i) Problem formulation:

Every study begins with clearly stating the problem. It is provided by policy makers and analyst it is clearly understood.

ii) Setting of Objectives plan:

It is the process of finding the way of approaching the problem. It includes the cost of study, no. of days required to accomplish each phase of work with anticipated results, a statement of the alternative systems and a method of evaluating their effectiveness.

iii) Model Conceptualization:

It is the process of establishing a reasonable model. It is the best to start with simple model and build toward greater complexity.

iv) Data Collection:

It is the process of collecting the data necessary to run the simulation such as arrival rate, arrival process, service rate, etc. Objective kind of data are to be collected.

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v) Model Translation: It is the process of converting a model into programming language.

vi) Verified: Here, we verify the model by checking if program works properly. If the input parameters and logical structure are correctly represented, verification is completed.

vii) Validate: Here, we check if the system accurately represents the real system and validate it.

viii) Experimental Design: How many runs? For how long? What kind of input variations?

ix) Production runs and analysis: It is the process of actually running the simulation and collect and analyze the input.

x) Repetition: Here, we repeat the experiments if necessary.

xi) Documentation and reporting: Here, we document and report the results. Two types of documentation: Program documentation which can be used to understand how program operates and Process documentation which gives history of simulation project.

xii) Implementation: If all the above steps are done with proper discipline, we implement the simulated system in the real world if the simulation results shows it is advantageous to implement the new system or policy.

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3. Describe different types of mathematical simulation models with suitable examples.

⇒ A mathematical model uses symbolic notation and mathematical equation to represent a system. The system attributes are represented by variables and the activities are represented by mathematical functions that interrelate the variables.

The different types of mathematical simulation models are given below:

1. Static mathematical model:

- A static mathematical model gives the relationship between the system attributes when the system is in equilibrium.
- If the point of equilibrium is changed by altering any of the attributes values, the model enables the new values for all the attributes to be derived but does not show the way in which they changed to their new values.

Example: Static Market Model.

- Supply and demand.

If P denotes Price, D denotes demand, S denotes supply.

$$\left.\begin{aligned} D &= a - bp \\ S &= c + dp \end{aligned}\right\} \quad \text{—— (1)}$$

$$\boxed{\therefore S = D}$$

a, b, c, d constants calculated based on previous market value.

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$S = D$

$$P = \frac{a-c}{b+d}$$

let, $a = 500$, $b = 2000$, $c = -50$, $d = 1500$

$$P = \frac{550}{3500} = 0.1571$$

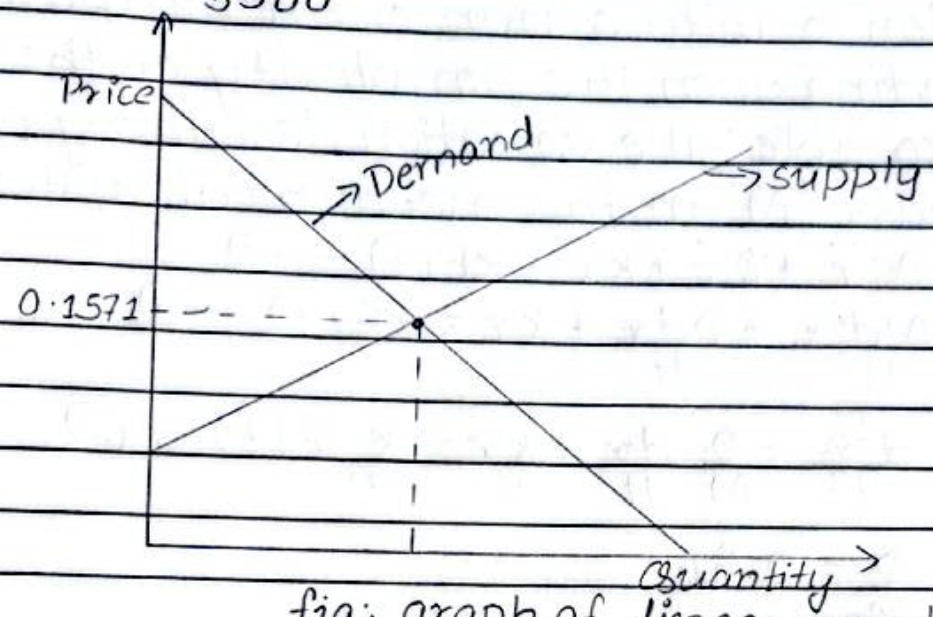

fig: graph of linear market model

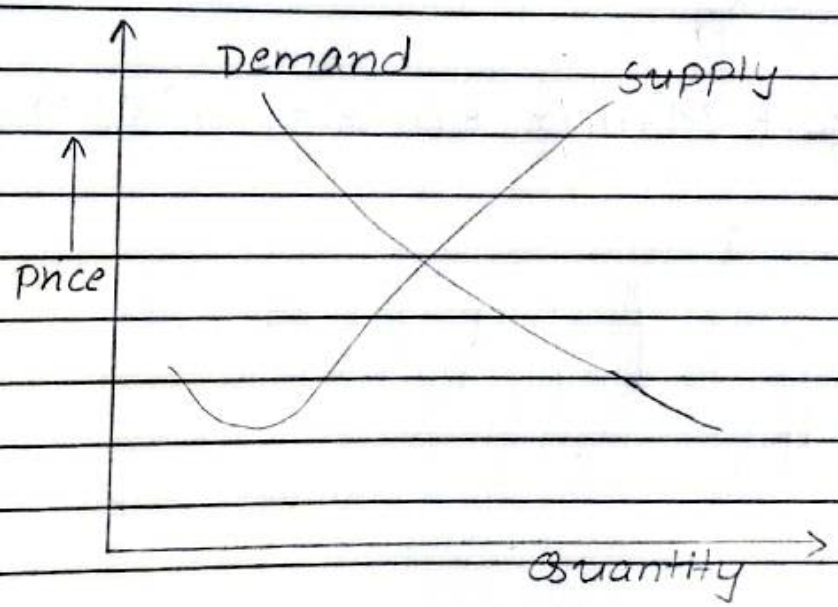

fig: graph of non-linear market model

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2 Dynamic Mathematical Model:

- A dynamic mathematical model allows the change of system attributes to be derived as a function of time.
- The derivation may be made with an analytic solution or with a numerical computation, depending upon the complexity of the model.

For example: the equation describing the motion of mass "M" in the mechanical system:

$$M\ddot{x} + D\dot{x} + Kx = KF(t) \quad \text{—①}$$

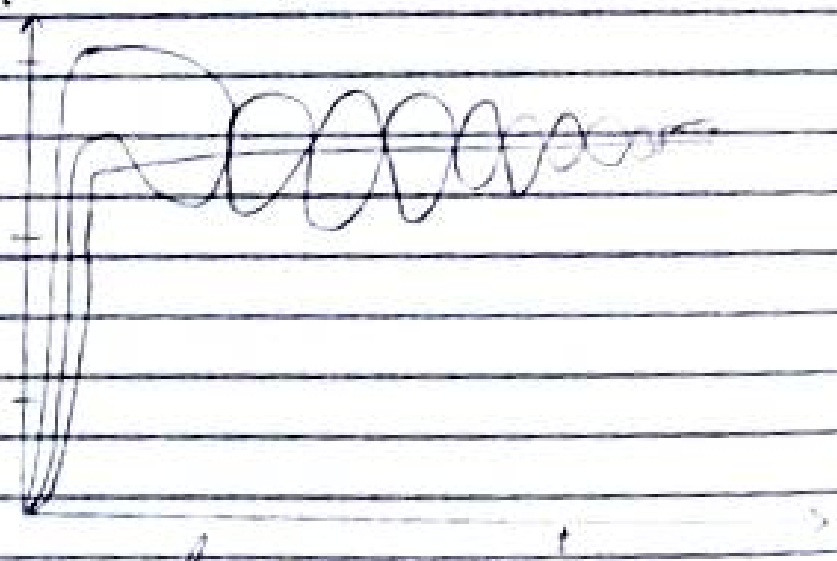
$$M\frac{d^2x}{dt^2} + D\frac{dx}{dt} + Kx = KF(t) \quad \text{—①}$$

$$\frac{d^2x}{dt^2} + \frac{D}{M}\frac{dx}{dt} + \frac{Kx}{m} = \frac{K}{m}F(t) \quad \text{—②}$$

$$\frac{D}{M} = 2\zeta\omega$$

$$\omega^2 = \frac{K}{m}$$

$$\frac{d^2x}{dt^2} + 2\zeta\omega\frac{dx}{dt} + \omega^2 x = \omega^2 F(t) \quad \text{—①}$$



$\zeta \geq 1$ motion in linear

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5. What is modeling? Explain different types of simulation model

⇒ Modeling can be used both as an analysis tool to predict the effect of changes to existing system and as a design tool to predict the performance of new system under varying set of circumstances.

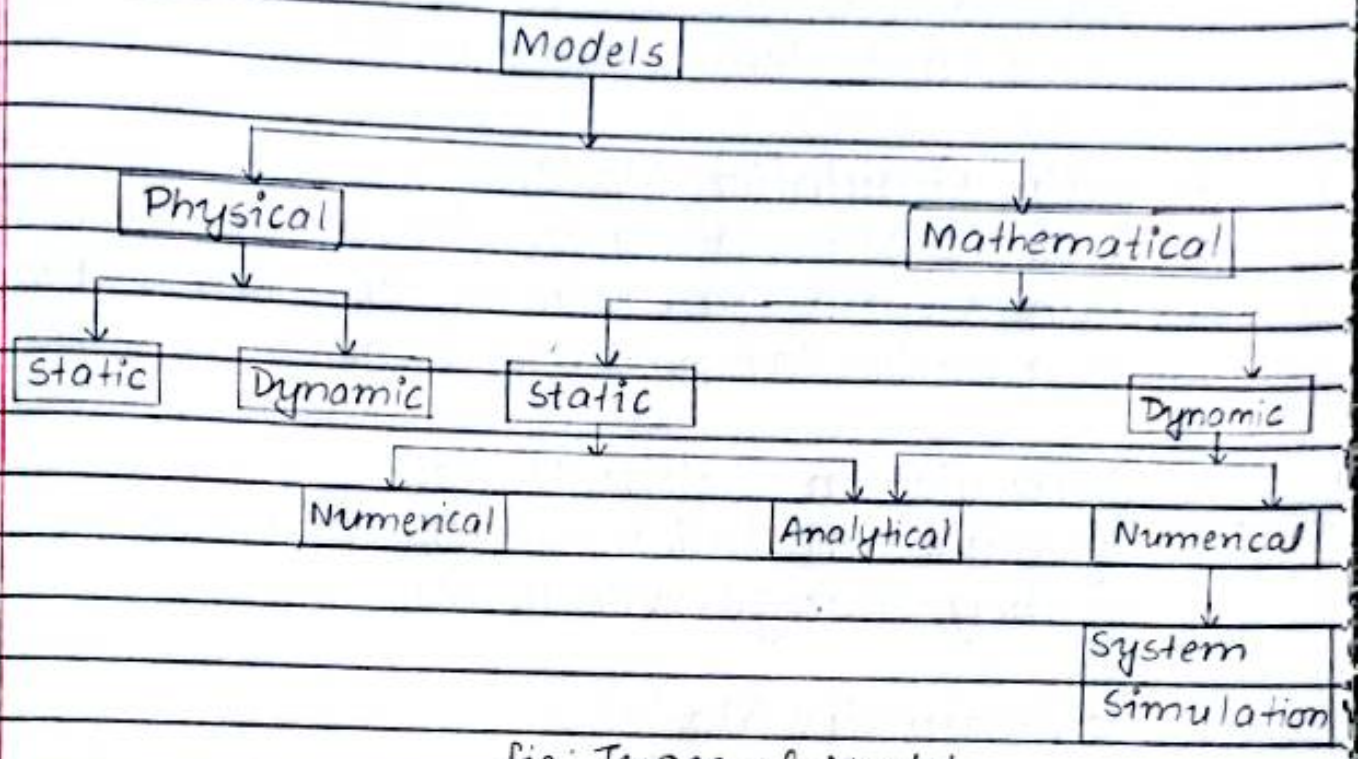

fig: Types of Model

1. Mathematical Model:

- A mathematical model uses symbolic notation and mathematical equations to represent a system.
- The system attributes are represented by variables and the activities are represented by mathematical functions that interrelate the variables.

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2. Physical Model:
- It is based on some analog between systems such as mechanical or electrical or hydraulic models.
- In a physical model of a system, the system attributes are represented by measurements such as a voltage or the position of shaft. The system activities are reflected in the physical laws that derive the model.

3. Static Simulation Model:
- A static simulation model, also called as Monte Carlo simulation, represents a system at a particular point in time.
-

4. Dynamic Simulation Model:
- Dynamic simulation model represent systems as they change over time.

5. Deterministic Model:
- Deterministic model have a known set of inputs which will result in a unique set of outputs.

6. Stochastic Simulation Model
- Stochastic Simulation model has one or more random variables as inputs. Random inputs lead to random outputs. Since the outputs are random they can be considered only as estimates of the true characteristics of a model.

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7. Analytical Model:

- These models use the deductive reasoning of mathematical theory to solve a problem. Once we built a mathematical model it must then be examined to see how it can be used to answer the question of interest about the system it supposed to represent.

8. Numerical Model:

- This model applies computational procedures to solve equations.
- Some analytic solution may require a considerable amount of computing resource. In such case, the model must be studied using numerical model.

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6. Explain about system and it's components with example.

⇒ System is simplified representation of selected components with specified boundaries and predefined time characteristics.

- It generally represents a real word entity over time and space in generalized and specified form.

The components of system are given below:

i) Entity: It is an object of interest in the system. Eg: In the factory system, departments, orders, parts and products are the entities.

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ii) Attribute: An attribute denotes the property of entity. Example: Quantities for each order, type of part, or number of machines in a Department are attributes of factory system

iii) Activity: Any process causing changes in a system is called as an activity. Example: Manufacturing process of the department.

iv) Event: An event is defined as an instantaneous occurrence that may change the state of the system. Examples: arrival of a new customer, completed service of a customer.

v) State variables: The state of a system is defined to be that collection of variables necessary to describe the system at any time, relative to the objectives of the study. Example: In the study of a bank, possible state variables are the number of busy tellers, the number of customers waiting in line or being served, and the arrival time of the next customer.

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System	Entities	Attribute	Activities	Events	State Variables
Banking	Customer	Checking account balance	Making deposits	Arrival; departure	Number of busy tellers; number of customers waiting
Production	Machine	Speed; capacity breakdown rate	Welding Stamping	Breakdown	Status of machines (busy, idle or breakdown

7. Explain system environment in simulation modeling.

⇒ The changes occuring outside the system that affect the system is called system environment. Defining a system boundary is important:

↳ Small system boundary may not include necessary components.

↳ larger system boundary may not have high degree of error propagation and management difficulties.

The different types of system environment in simulation modeling are given below:

i) Endogenous System:

The term endogenous is used to describe activities and events occuring within a system

Example: Drawing cash in a bank.

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ii) Exogenous System:
The term exogenous is used to describe activities and events in the environment that affect the system. Example: Arrival of customer

iii) Closed System:
A system for which there is no exogenous activity and event is said to be a closed. Example: Water in an insulated flask.

iv) Open System:
A system for which there is exogenous activity and event is said to be a open. Example: Bank system.

8. What do you mean by distributed lagged model? explain it?

⇒ The model that have properties of changing only at fixed interval is known as distributed lagged model.

For example: Mathematical model of national economy.

Let, C = consumption
I = Investment
T = Taxes
G = Government expenditures
Y = National income.

Then,

$$\left.\begin{aligned} C &= 20 + 0.7(Y - T) \\ I &= 2 + 0.1Y \\ T &= 0.2Y \\ Y &= C + I + G \end{aligned}\right\} \text{——— (1)}$$

All quantities are expressed in billions of rupees. This is static model and can be made dynamic by lagging all the variables as follows:

$$C_i = 20 + 0.7(Y_{i-1} - T_{i-1})$$
$$I_i = 2 + 0.1Y_{i-1}$$
$$T_i = 0.2Y_{i-1}$$
$$Y = C_{i-1} + I_{i-1} + G_{i-1}$$

Any variable that can be expressed in the form of its current value and one or more previous value is called lagging variable. And hence, this model is given the name distributed lag model.

- Advantages of distributed lag model:

i) Simple to understand and can be computed by hand, computers are extensively used to run them.

ii) There is no need for special programming language to organize simulation task.

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9. Explain different phases in simulation model with flow chart.

⇒ The different phases in simulation model are given below:

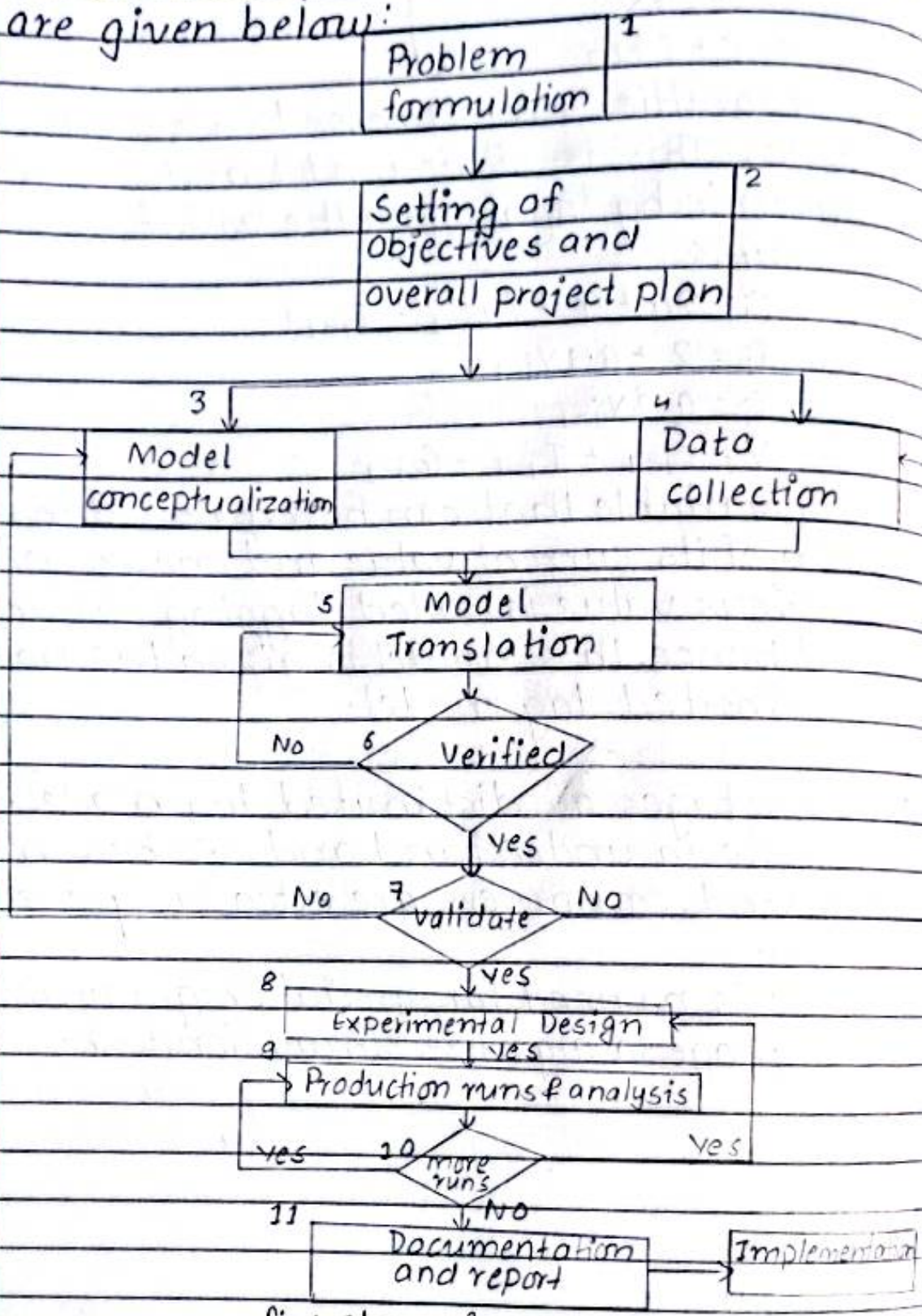

fig: steps in simulation modeling

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All these steps can be divided into four phases:

a) I phase (discovery phase):
- Consists of steps 1 and 2.
- It is period of discovery/orientation.
- The analyst may have to restart the process if it is not fine-tuned.
- Recalibration and clarifications may occur in this phase or another phase.

b) II phase (intermediate phase):
- Consists of steps 3, 4, 5, 6 and 7.
- A continuing interplay is required among the steps.
- Exclusion of model user results in implications during implementation.

c) III phase (experiment phase):
- Consists of steps 8, 9 and 10.
- Conceives a through plan for experiments.
- Proper statistical analysis is required.

d) IV phase (Implement phase):
- Consists of step 11 and 12.
- Successful implementation depends on user and every steps successful implementation.

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12. Explain applications of simulation.

⇒ The applications of simulation are given below:

a) Manufacturing Application:
- Analysis of electronics assembly operations.
- Optimization of cycle time and utilization in second conductor test manufacturing.
- Analysis of storage and retrival strategies in a warehouse.

b) Construction Engineering:
- Construction of a dam embankment.
- Activity scheduling in dynamic, multiproject setting.

c) Military Applications:
- Modeling military requirements for non-war fighting operations.
- Design and test of intelligent controllers.

d) Human System:
- Modeling human performance in complex systems.
- Studying the human element in traffic control.

e) Business Process Simulation:
- Product development program planning.
- Reconcillation of business and system modeling.

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13. Explain advantages of simulation.

⇒ The advantages of simulation are given below:

i) Simulation can also be used to study systems in the design state.

ii) Simulation models are run rather than solved.

iii) New designs can be tested without acquiring resources.

iv) New procedure or rules can be explored without hampering the ongoing real time system operations.

v) Insights can be obtained about the variable interaction and their credit for system performance.

vi) Insight can be obtained about the interaction of variables.

vii) The hypothesis about certain phenomena can be tested for feasibility.

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14. Explain real ~~type~~ time simulation.

⇒ In real time simulation, actual device (which are part of a system) are used in conjuction with either digital computer or hybrid computer. It provides the simulation of the points of system that don't exist or that cannot be easily used in an experiment i.e the basic idea of real time simulation is, uses the actual part if they are appropriate

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to use in experiment otherwise use the simulation of the points of the system. A well-known example is "simulation to train pilots". It uses the devices for training pilots by giving them the impression that is at the control of an aircraft. It requires real time simulator of the plane its control system, the weather and other environmental conditions.

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